Amith Polineni Project Portfolio

Hello! I'm Amith Polineni. This portfolio showcases several projects I have worked on, highlighting my experience in robotics, mechanical design, automation, control systems, and human spaceflight systems integration at NASA. Each project demonstrates my ability to combine engineering creativity with technical expertise. You can explore these projects in more detail on my website: amithp.com.

NASA HQ Strategic Integration (Pages 1–3)

- Served as the technical arm of NASA HQ's Strategic Integration and Management Division (SIMD).
- Architected a unified SharePoint process handbook: 24 mission-support processes, 6 categories, built while SOMD and ESDMD merged.
- Ran a 9-week program from process inventory and SME interviews to UX mockups, build, and refactor.
- Gained JSC access (Mission Control, Neutral Buoyancy Lab, HERA) and attended the Artemis III crew announcement.
- Represented NASA for 4 days at the Great American State Fair.

Figure: NASA Headquarters

Robotic Stroller Research (Pages 3 & 4)

- Engineered a front-following robotic stroller by integrating stereo vision, LiDAR perception, and ROS-based control.
- Developed pose-tracking, dynamic replanning, SLAM, and PID motion smoothing for real-time navigation.
- Optimized mechanical systems with Onshape CAD and 3D printing, including a differential gearbox and reinforced steering.
- Applied Bayesian control logic for reliable autonomous operation at outdoor speeds.
- Designed modular electronics with a 24V network, buck converters, and safety interlocks.

Figure: Robotic Stroller

MARS Robotics: 1st Place & 2x Innovation Award (Pages 5–9)

- As VP and design lead, took Bruno to 1st overall and the Off-World Grand Prize at NASA Lunabotics, plus a second straight Innovation Award (Artie: 5th, 2025).
- Designed from first principles around a ConOps that saves 20 s per cycle: 170+ L in 15 minutes on 23.7 Wh.
- Lost a wheel mid-run at KSC and still won, driving on the robot's own drums.
- Owned chassis, arm, dig-and-dump, and e-box mechanical design, plus e-box assembly.

Figure: Bruno at NASA Lunabotics

Automatic Aquaponic Farm (Page 10)

- Designed an IoT-enabled aquaponics system using a Raspberry Pi with web-based monitoring.
- Automated irrigation and lighting for a 10-channel vertical farming system, optimizing efficiency.
- Operated autonomously for 10 weeks, growing and harvesting kale, lettuce, Swiss chard, and spinach.
- Applied rapid prototyping and low-cost solutions, reducing expenses by 80%.

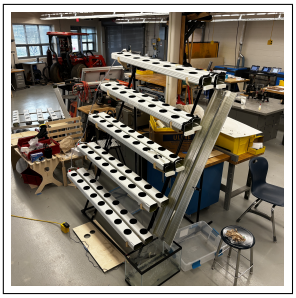
Figure: Aquaponic Farm

1 NASA HQ Strategic Integration & Management Division

OSTEM Intern, Human Spaceflight Mission Directorate, NASA Headquarters (Summer 2026)

Human spaceflight only works when thousands of moving parts move together, and the **Strategic Integration and Management Division (SIMD)** is the team that makes that happen. I joined SIMD as the **technical arm** of that effort: I owned a summer-long engineering deliverable from first inventory to final demo, while the directorate was merging two organizations (**SOMD** and **ESDMD**) underneath the project. The result was a unified, searchable process handbook, plus a front-row seat to how NASA actually operates at mission scale.

1.1 The Problem: Scattered Processes, Shifting Ground

Mission-support processes were everywhere and nowhere. Procedures for **DPMC**, **Contracts & Procurement**, **CUI Compliance**, **Risk Management**, **IT Equipment**, **STRIVES**, **Awards & Recognition**, and **Onboarding** were spread across many platforms, with no single place to find an owner, a form, or a next step.

The reorganization of SOMD and ESDMD made the target move mid-project: processes changed, and so did SIMD's client group. I treated that as a systems-integration requirement rather than a setback. The handbook's purpose evolved from convenient reference to a tool that **helps align merging processes**.

1.2 The Solution: One Handbook, One Source of Truth

Twenty-four processes. Six categories. One front door. I designed and built the handbook as a **SharePoint-based site** organized into Administration & Staffing, Communications & Publications, Digital Services & Cybersecurity, Governance & Councils, Risk, Knowledge & Records, and Program Formulation & Acquisition.

Every process page follows the same template, so a reader can find what they need in seconds:

- **Brief intro**: what the process is and why it exists.
- **External resources**: authoritative forms, toolkits, and SharePoint links.
- **Points of contact**: named process leads, with roles.
- **Detailed process**: a step-by-step walkthrough.
- **Left-hand navigation**: category-level browsing across the whole handbook.

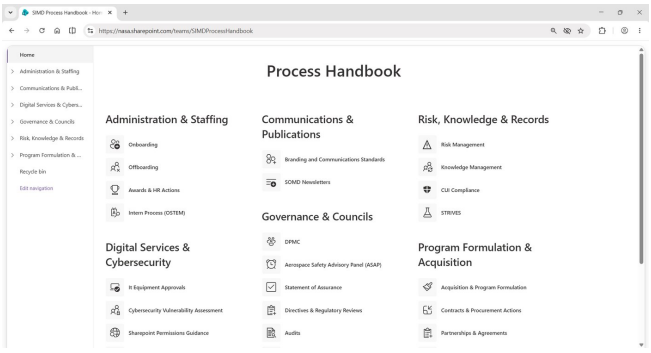

Figure 1: Handbook home page: six categories, 24 processes.

Figure 2: Anatomy of a standardized process page.

1.3 Engineering the Program

I ran the project as two parallel workstreams: **Handbook Site** (brainstorm layouts, mock up, populate with real processes, refactor and polish) and **Process Management** (build the process inventory, interview subject-matter experts, run additional SME interviews). Four milestones kept it honest: an initial check-in, a visual-layout demo meeting, a handbook feedback meeting, and a final presentation. The schedule flexed around two **Johnson Space Center trips** and the Great American State Fair.

1.4 Process Inventory and Documentation Audit

Before building anything, I stood up a **process inventory** as a SharePoint list, tracking each process's owner, whether documentation already existed, and consolidation progress. It doubled as an audit. Documentation quality varied widely:

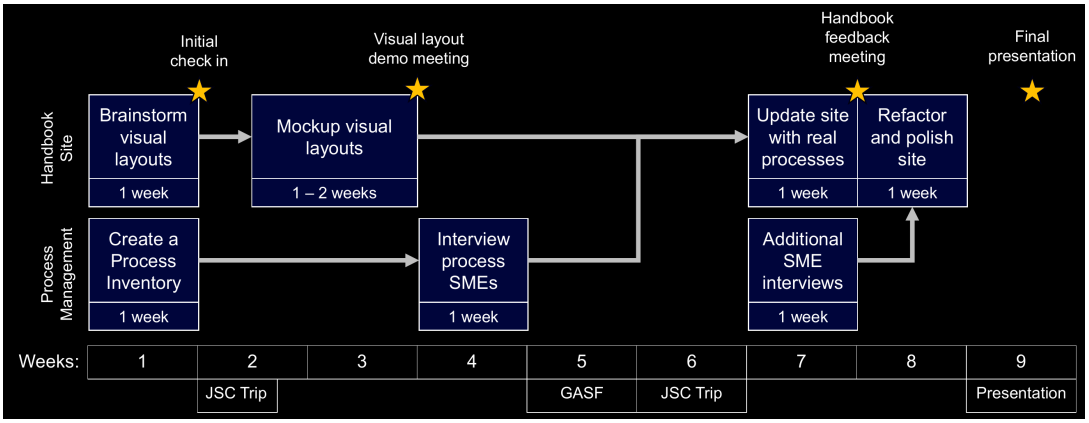

Figure 3: Nine-week project timeline with two parallel workstreams and four milestone reviews (stars).

some processes were **fully documented**, many only **partially**, some **not at all**, and some were **dropped** entirely. For the undocumented ones, **SME input was essential**. I interviewed process leads to capture knowledge that existed only in people's heads.

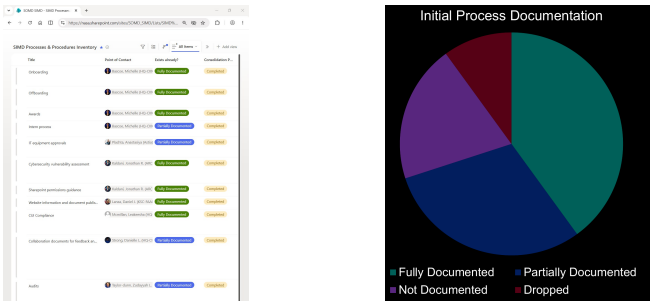

Figure 4: Process inventory (left) and initial documentation status across all processes (right).

1.5 Refactoring for Editability

My first build optimized for usability alone, which made adding a new process painful. I **refactored the site architecture** so new processes drop in cleanly, with no loss of usability. The handbook is designed to keep growing after my last day. Keeping the inventory current paid off too: it caught processes I had missed and gave me a clean way to manage processes being added and removed. (Also verified in the field: Chrome beats Firefox for SharePoint.)

1.6 Strategic Communications and JSC Access

Alongside the handbook, I supported strategic communications and got access that few interns see:

- **Johnson Space Center**: Mission Control Center, Neutral Buoyancy Lab, Space Vehicle Mockup Facility, Systems Engineering Simulator, HERA, and Space Center Houston, plus the **Artemis III crew announcement**.
- **Great American State Fair**: four days at the NASA booth, including the July 5th crowds. I briefed the public on **Artemis and the Moon Base**, learned the programs straight from NASA SMEs, and was featured on **Jared Isaacman's Instagram**.
- **JSC Silver Snoopy Awards**: supported the JSC Space Flight Awareness (SFA) team, owned recipient check-in, and met astronaut **Stephen Bowen**.
- **SFA inventory and ISS astronaut display**: counted and organized pins, patches, and decals spanning STS missions, ISS expeditions, and Dragon crews, and helped update the HQ hallway display and prepare a new HQ lobby display.

Figure 5: Staffing the NASA booth at the Great American State Fair.

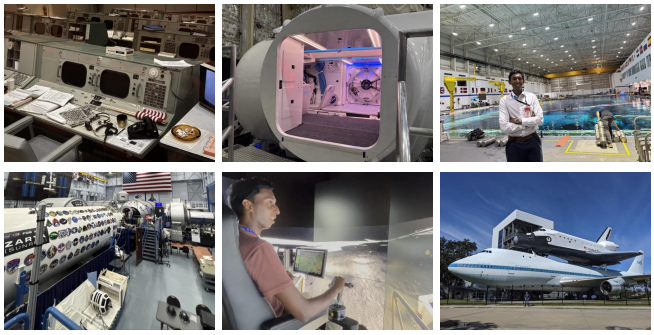

Figure 6: JSC access: Mission Control, Neutral Buoyancy Lab, Space Vehicle Mockup Facility, Systems Engineering Simulator, HERA, and Space Center Houston.

1.7 Takeaways

Working across so many mission-support areas made one thing clear: the heavy oversight in human spaceflight is **proportional to mission scale**, and every process has a clear lead if you know where to look. The most impactful parts of the summer were the **feedback meetings**, the **trips to JSC**, and the **Great American State Fair**. NASA's culture is exceptional. I left with a deliverable that outlives the internship, and an application to **NASA Pathways** already on my list.

Figure 7: Silver Snoopy Awards with astronaut Stephen Bowen (left) and inside an Orion mockup at JSC (right).

2 Robotic Stroller Research

2.1 Motivation

This project focuses on developing a **front-following robotic stroller** designed to operate in outdoor environments at jogging speed. Unlike traditional follow-bots, which trail behind a user, this stroller positions itself in front. The core innovation is using body pose estimation to infer human intent and generate predictive navigation trajectories.
Front-following provides critical benefits:

- The robot remains visible to the user, increasing trust and predictability.
- Elderly or visually impaired users can be guided more intuitively, as the stroller acts as a visible lead rather than an unseen follower.
- In crowded sidewalks, the stroller actively clears a path, rather than weaving through obstacles reactively.

This capability required fusing **stereo vision (ZED/ZED2i)**, **LiDAR sensing**, and **ROS-based control** to perform real-time SLAM, obstacle avoidance, and trajectory generation.

Figure 8: Overview of the stroller design with rear-wheel drive and front caster steering.

2.2 Mechanical Reinforcements

Initial testing revealed reliability issues in the drivetrain. The bevel gears of the differential gearbox slipped under sudden torque spikes, caused by sharp accelerations or uneven outdoor terrain. This not only produced noise but also risked long-term structural damage.
To address this, I CAD-modeled custom bracing mounts in Onshape and 3D-printed reinforcements that clamped around the gearbox housing. These reduced flex in the support structure and kept bevel gears in tight alignment. In practice, this eliminated gear skipping and allowed the stroller to sus-

tain higher speeds with confidence.

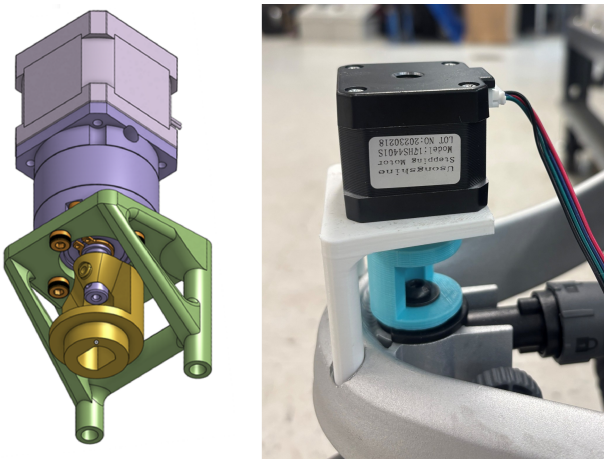

Figure 9: CAD of upgraded gearbox mount (left) and real-life support piece (right).

2.3 Steering and Sensing Upgrades

The front caster wheel posed a unique challenge: by default, caster wheels self-align forward, creating large resistive forces during sharp turns. To overcome this, I integrated a **high-torque motor with a planetary gearbox**, enabling the robot to hold stable steering angles even under caster-induced torque. Thicker mounting brackets were added to prevent deflection at high speeds.
The Ouster OS1 LiDAR was mounted using vibration-dampening 8020 slot inserts. These reduced jitter in point cloud data, ensuring consistent obstacle detection while the stroller navigated over sidewalks and curbs.

2.4 Human Following Control

The stroller's human-following behavior was implemented in ROS with multiple control layers:

- **Pose estimation**: using ZED stereo cameras for outdoor reliability, supplemented by Mediapipe on an Intel RealSense camera for shoulder landmark extraction.
- **Trajectory generation**: computing a forward offset vector from the user's shoulders, then replanning dynamically around obstacles with SLAM-informed maps.
- **Motion smoothing**: raw commands caused jerky motion, so low-pass filters and PID controllers were added to smooth both translational and angular velocities.
- **Joystick override**: a ROS teleop node was integrated for manual override, blending joystick and autonomous inputs for safety.

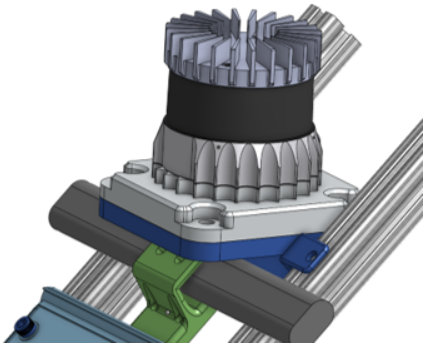

Figure 10: OS1 Lidar sensor mounted in CAD

2.5 Pose Estimation for Direction Prediction

Human orientation was estimated by tracking shoulder vectors. By computing a normal vector across the shoulders, the stroller could infer where the user intended to go, even before they stepped forward. This predictive behavior gave the robot more fluid, human-like motion, rather than reacting late to position changes.
Mediapipe experiments showed that even lightweight models could reliably track these landmarks outdoors. Combined with ZED's depth estimation, the stroller maintained a safe lead distance of 1.5–2 meters at running speed.

Figure 11: Pose estimation with orientation vector derived from shoulder alignment.

2.6 Planning and Safety Algorithms

A dynamic **around-person planner** was implemented, treating the human as a moving reference point. The stroller generated paths that stayed in front of the user while respecting:

- A circular "safety bubble" around the user,
- Velocity and acceleration limits for stability,
- Obstacle avoidance constraints from LiDAR.

This approach allowed the robot to maintain a leading position while smoothly navigating crowded or cluttered paths.

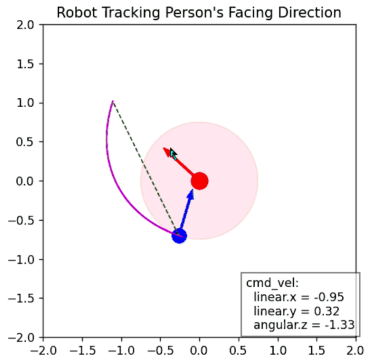

Figure 12: Simulation of robot maintaining position in front of a moving person.

2.7 Electronics and Power System

The stroller's electronics were designed for modularity and fault tolerance. Key features include:

- A 6S LiPo battery with **buck converters** supplying 24V, 14.8V, and 5V rails for motors, sensors, and logic.
- A distributed microcontroller system handling low-level sensor integration and motor actuation.
- A ROS laptop controller for high-level planning, communication, and visualization.
- **Emergency stop circuitry** with hardware interlocks to cut power instantly in case of fault.

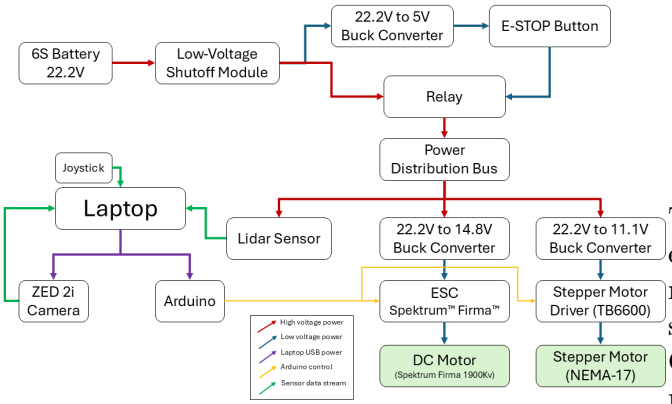

Figure 13: Electronics layout including buck converters, distributed controllers, and emergency stop system.

2.8 Outcomes and Future Work

The reinforced gearbox and steering upgrades allowed reliable high-speed outdoor operation. Pose-estimation-based leading made the stroller's behavior intuitive and trustworthy, while smoothing filters and PID tuning improved fluidity.

3 MARS Robotics: Artie, 5th Place & Innovation Award (2024–25)

In 2024–25, our Mechatronics and Robotics Society (MARS) team of 40 members took on NASA's Lunabotics lunar excavation challenge. As a mechanical team member and design lead, I proposed a radical concept for a bucket drum excavation system. After multiple rounds of design reviews and trade studies, my concept was selected as the official robot design: **Artie**. Artie was deployed at NASA's competition, where it placed **5th out of 70 teams** and won the **Innovation Award**. That performance led to my promotion to Vice President the following year, leading mechanical design, systems integration, and performance optimization for the team (see Bruno, next section).

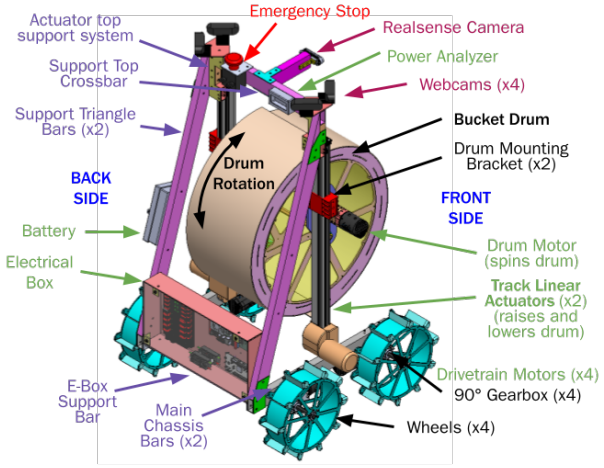

Figure 14: Labeled final configuration of robot

The bucket drum robot was inspired by advanced extraterrestrial excavation systems. It employs a rotating, baffled drum with strategically placed scoops to efficiently collect and transport regolith. Our goal was to maximize material retention, optimize torque distribution across the drum, and design a robust system capable of high reliability under dynamic operational conditions.

3.1 Preliminary Design Stage

During the initial design stage, the concept featured a dual drum configuration mounted on linear actuators, providing adjustable deployment height for excavation and transport. Segmented drum scoops were staggered to evenly distribute torque and reduce peak motor loads. The early focus was on proof-of-concept, with simplified mechanical modeling to validate functionality, actuation kinematics, and overall system feasibility. Real commercial ac-

tuators were used to ensure realistic motion limits and inform future integration and control strategies.

3.2 Intermediate Design Stage

By the intermediate design stage, the dual drum design was refined to include three regolith collection openings per drum to accelerate excavation cycles during short operational runs. Flexible thermoplastic living hinge flaps were explored to retain material, but ultimately removed to improve reliability and reduce mechanical complexity.
Structural optimization was a major focus: triangular support braces were implemented to transfer actuator loads directly to the chassis, reducing stress on individual components. The electronics enclosure was repositioned for accessibility and modular wiring, and critical joints were reinforced with gussets and precision hardware. Half-scale prototypes revealed that internal baffling was essential to retain material during drum rotation, prompting a redesign of drum geometry with two large openings to maximize storage per cycle.

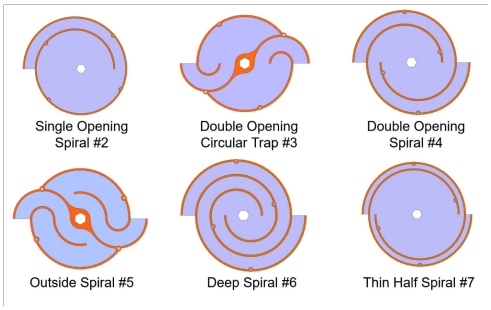

Figure 15: Variations on bucket drum geometry

3.2.1 Drum Geometry and Rapid Prototyping

We conducted extensive rapid prototyping with 3D-printed subscale drums to test internal baffling and opening configurations. Designs included single-spiral, double-spiral, and circular trap geometries to analyze regolith flow dynamics, material retention, and torque distribution. Iterative testing allowed us to optimize baffle placement, opening size, and internal volume utilization while minimizing weight and simplifying manufacturing. These tests confirmed that practical fill capacity was limited to approximately half the drum volume, guiding the final drum geometry.

Figure 16: Subscale prototypes of potential bucket drum geometries with annotations after testing

3.3 Critical Design Stage

For the final design, the drum was scaled to the maximum allowable dimensions to optimize material capacity while remaining within competition constraints. A single-drum design enabled continuous sheet metal fabrication, improving structural integrity and manufacturability. Interlocking finger joints facilitated precise assembly and weldability. Track actuators were upgraded to larger, high-load-capacity units, reducing the number of components, simplifying control, and increasing dynamic load tolerance. Structural supports were refined, with redundant braces removed and remaining elements strategically reinforced to minimize weight without compromising rigidity.

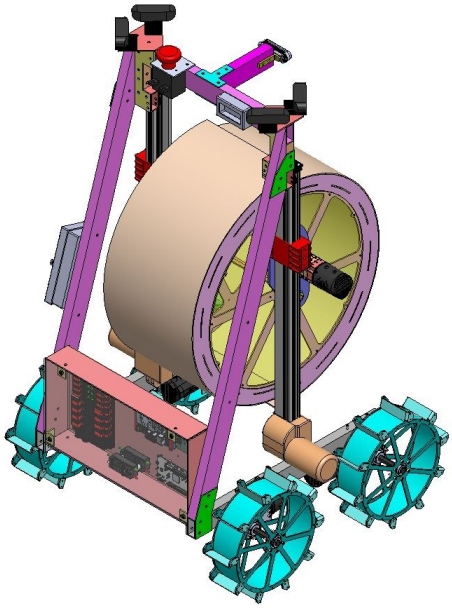

Figure 17: CDR (Feb) bucket drum robot CAD

3.4 Post-Manufacturing Optimization

Extensive field testing highlighted high torque demands, leading to replacement of the original belt-and-pulley system with a direct-drive motor coupled via a bevel gearbox. To meet peak torque requirements, a second motor was integrated, ensuring consistent rotational speed under load. Flexible polycarbonate side panels were reinforced with adhesive and metallic tape to prevent regolith spillage.

Dust protection for gearboxes and exposed actuators was enhanced, and camera mounts were redesigned as modular, lightweight components to allow rapid adjustment and improve sensor coverage.

Figure 18: Bucket Drum robot at sand court testing

Weight reduction and manufacturability were further addressed by pocketing non-load-bearing sections of structural supports, refining top-frame geometry, and consolidating components. This iterative approach to subsystem integration, torque management, and geometric optimization resulted in a high-performance, reliable bucket drum robot capable of excelling in a competitive lunar excavation environment.

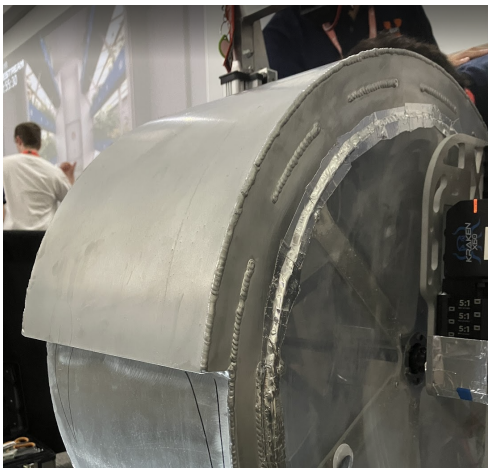

Figure 19: Drum with Aluminum Foil Tape seal

Figure 20: In the regolith area with the robot (I'm the one on the right)

4 MARS Robotics: Bruno, 1st Place & Back-to-Back Innovation Awards (2025–26)

Artie's 5th place finish set the bar, and this year the goal was to **win the whole competition**. As **Vice President and design lead** of a 105-member MARS team, I led the mechanical design of **Bruno**, a dual-drum lunar excavation and construction robot. At NASA's Lunabotics Challenge, Bruno finished **1st overall** and took home the **Off-World Grand Prize** for UVA, plus the **Innovation Award**, 2nd in robotic construction, 2nd in systems engineering, 2nd in autonomy, and special recognition for mission excellence. That is six awards at Kennedy Space Center, and a jump from 5th place to 1st in a single year.

NASA Lunabotics Innovation Award: two years running

2025: Artie, 5th place · 2026: Bruno, 1st place and Grand Prize

Two different robots, two different architectures, and the same result from NASA's judges. Back-to-back Innovation Awards are the clearest signal that the design philosophy behind both machines, starting from the physics of the task rather than from last year's robot, works.

4.1 Designing from First Principles

Only one constraint was fixed: the bucket drum. Everything else was derived. I started from what limited Artie's performance: one large central drum (11.5 in wide, 25 in diameter) raised the center of gravity, so we drove conservatively to avoid tipping, and that caution cost us cycles and points. Bruno splits the excavation mechanism into **two pairs of smaller drums, one pair on each side of the chassis**. This lowers and widens the mass distribution, so the robot can traverse craters and boulders aggressively. It also gives a **56-inch cutting edge, roughly 5× Artie's**, and each arm lowers independently, so we can tune the center of mass on the fly. A single large drum on a pivoting arm was a serious alternative, but its cantilevered structure and control complexity lost the trade study to the symmetric layout.

The design targets came straight from a scoring analysis: system mass at or below 35 kg (Artie was 43.5 kg), at least 6 L/min of regolith transfer, and average power at or below 180 W. We kept a square wheelbase for in-place turning, and built a rough full-scale prototype three months ahead of schedule from scrap stock and reused parts. It was so sound that the original chassis and arms made it all the way to competition.

4.2 Concept of Operations: Winning in the Timeline

A lunar excavation robot is a timing problem as much as a mechanical one. I designed Bruno around a **four-phase cycle**: excavate (60 s), traverse (60 s each way), and dump (60 s), for a 4-minute cycle and four cycles in a 15-minute run.

The most impactful decision was **raising the arms while driving**. The arms need roughly 20 s to lift, so they rise concurrently with the 60 s drive across the obstacle zone, saving **20 seconds every cycle**. The drums sit on opposite sides, so deposition is a choreographed sequence: dump one side, back up, turn 180 degrees, dump the other. The first cycle is autonomous, and works reliably because the arms' starting configuration is fully predictable.

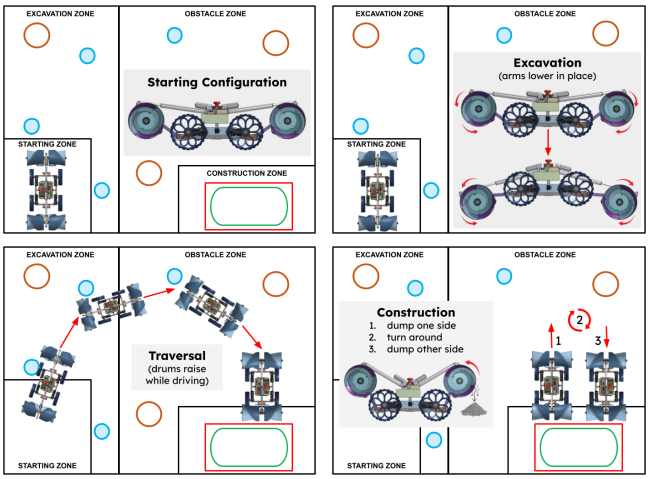

Figure 21: Bruno's ConOps: start, excavate in place, traverse with arms rising, and dump one side then turn to dump the other.

4.3 Digging and Dumping Mechanics

The drum mechanism is **static and bidirectional**. Spin the drums one way and they collect regolith, reverse them and the same openings deposit it, so the robot excavates *and* constructs the berm with no extra moving parts to hold or transfer material. Linear actuators with hard stops lower the arms to a repeatable depth (0.10 m below the wheel contact plane), which makes digging consistent from cycle to cycle. Clear polycarbonate side panels let the operator read drum fill level directly through a situational-awareness camera. Unit tests of the cantilevered arm showed the full-drum lift load exceeded the original motor's capability, so I re-architected the lift to a more robust arm drive. In integrated testing, the drums regularly collected more than the theoretical 50% fill.

4.4 Motor Current and Energy Optimization

Lunabotics scores energy and mass productivity, so **every watt is points**. The previous year's robot used 18 A motor current limits to protect the compute, which caused stalling. For Bruno, I treated motor current as a mechanical design constraint: low-inertia drums that spin up easily, arms driven by linear actuators with hard stops, and ramped drum-motor stops so current and voltage spikes stay low.

The payoff was measured: in our second integrated system test, a simulated 15-minute run deposited **over 170 L of regolith on only 23.7 Wh**. That works out to roughly 11 L/min against a 6 L/min target, at an average draw of about 95 W against a 180 W budget.

4.5 Manufacturing and the E-Box

Beyond the design, I built Bruno. I handled the **mechanical manufacturing and assembly of the electronics box (e-box)**, the robot's sealed brain: a regolith-proof housing, an accessible layout for the Jetson, power distribution, and motor-control electronics, and an e-stop mounted where the competition rules demand it. We 3D-printed the housing to iterate quickly, and every part was labeled so Bruno could be disassembled for the 800-mile drive to Florida and rebuilt, rewired, and verified in one night.

Figure 22: Reassembling Bruno on arrival in Florida, hours before competition.

4.6 Competition: The Wheel Fell Off. We Still Won.

We went into the NASA competition at UCF's Exolith Lab as one of 47 teams, and Bruno **won the qualifier with almost 50% more points than second place**, earning a spot among the top ten in the Kennedy Space Center finals.

Then, in our first KSC run, **a wheel failed and came off mid-run**. For most robots, that ends the run. Bruno's dual-drum layout has a hidden capability: lower the drums until the wheels lift off the ground, spin both drum sets in the same direction, and the drums become the drivetrain. We had practiced this **drum-traversal** mode before leaving Charlottesville. The remaining three wheels could not cross the terrain, but the drums could, and Bruno kept moving and scoring for the rest of the run. Our packing discipline paid off too: we had spares for everything, including a 3D-printed spare wheel that rode down to Florida in its own minivan.

Figure 23: Bruno recovering from a run-ending failure by driving on its drums, on the KSC competition livestream.

The result: despite losing a wheel, we placed **2nd in robotic construction** and won the **overall competition and the Off-World Grand Prize**, along with the Innovation Award for the second consecutive year. The lesson was precise: 3D printing is a superpower for iteration but critical, load-bearing parts deserve proven materials. The engineering takeaway is the one I'd defend in any design review: build for the failure you did not predict, and practice recovering from it.

5 Automatic Aquaponic Farm in a Greenhouse

This project involved the design and implementation of a custom Internet of Things (IoT) aquaponics system. The system was engineered to facilitate autonomous growth of leafy greens using a Nutrient Film Technique (NFT) framework, optimized for space efficiency and remote control. The primary objective was to leverage automation and cost-effective solutions to create a viable, sustainable, and remotely monitored agricultural system.

Figure 24: Me with the final prototype of the farm

5.1 System Architecture

The aquaponics system was designed as a 10-channel vertical farm, built to fit within a $10\times3\times6$ ft space, though the final implementation used 8 channels to accommodate greenhouse space limitations. The skeleton was constructed from 2020 aluminum extrusions. Components like channel supports were created using a hybrid manufacturing approach: 3D printing for complex, custom brackets and laser cutting for flat components, which proved to be faster and more cost-effective.

The system utilized rectangular vinyl downpipes instead of the traditional round PVC pipes. These 3-inch by 4-inch downpipes were tilted at a 7 degree angle to ensure proper water flow and prevent pooling. Plants were grown in net cups filled with clay pebbles.

Water circulation was managed by a pump and a network of tubes. To address the issue of inconsistent flow and prevent lower channels from receiving all the water, we implemented barb valves at each level to equalize pressure across all channels. A main collection channel with bent chutes and funnels was designed to direct falling water back into the main tank, although minor leakage was still an issue.

IoT and Automation

A Raspberry Pi served as the central control unit for the system's automation, integrated via a relay circuit. The Raspberry Pi was programmed to automate the water pump, operating on a 12-hour cycle (7 am to 7 pm) to provide nutrients to the plants. It also controlled automated grow lights to supplement natural light.

The system was designed to connect to a web server via a websocket for remote monitoring, enabling real-time data collection and display of environmental parameters such as temperature and humidity. The Raspberry Pi was programmed to send email notifications if the system lost connection with the server.

The project successfully demonstrated the feasibility of a remotely monitored and automated hydroponics system. The system was able to operate autonomously for 10 weeks, successfully growing kale and spinach.

Figure 25: Kale growing in the greenhouse

The frame and NFT channel design proved structurally sound and effective for plant growth. The autonomous pumping and lighting cycles were operational, fulfilling the core automation objective.

The project validated the potential of IoT-enabled vertical farming and provided valuable insight into the challenges of implementing such systems in a real-world environment.